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M-009 METERING PUMPS

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M 009 METERING PUMPS

1. SCOPE OF WORK

The Bidder shall supply the following in accordance with Part 4, Subsection 1.2. “Summary of Works”.

The pumps for plant shall include, but not be limited to the pumps included in Clause 3 herein and its associated electrical equipment as detailed in Clause 8 herein.

Metering pumps specified else where in Section 3 Technical Specifications or as required by the Bidder shall be subject to the design, testing and quality assurance requirements of this specification as applicable.

2. CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4, Subsection 2.5 “Codes & Standards”.

The Codes and Standards of the following organizations extracted from Subsection 2.5 are specifically applicable to the design, manufacture and testing of the equipment.

3. DESIGN AND OPERATING CONDITIONS

PUMP DESCRIPTION

All the following data shall be submitted by the Bidder for each metering pumps but not limited to:

SPECIFICATION SCOPE		ITEM 1	ITEM 2
3.1	Equipment Name..... :	Demineralization acid Regenerant metering pumps	Demineralization Caus- tic Regenerant meter- ing pumps
3.2	Quantity..... :	2 x 100%	2 x 100%
3.3	Location..... :	Water treatment area	Water treatment area
3.4	Liquid to be handled..... :	Hydrochloric acid	Sodium hydroxide
3.5	Temp. of liquid.....(°C) :	20 – 40	20 – 40
3.6	Pump suction from..... :	Acid storage tank	Caustic storage tank
3.7	Pump discharge to..... :	Mixed bed units	Mixed bed units
3.8	Pump capacity.....(l/s) :	By Bidder -ABO	By Bidder -ABO
3.9	Pump suction head.....(m) :	By Bidder -ABO	By Bidder -ABO
3.10	Total head.....(m) :	By Bidder -ABO	By Bidder -ABO
3.11	Material*		
3.11.1	Casing..... :	-CD -ABO	-CD -ABO
3.11.2	Diaphragms..... :	-CD -ABO	-CD -ABO
3.12	Flange		
3.12.1	Suction flange rating..... :	By Bidder -ABO	By Bidder -ABO
3.12.2	Discharge flange rating (ANSI std) :	By Bidder -ABO	By Bidder -ABO

* The Bidder must ensure that all material selected are suitable for the service required.

- CD = Bidder to Determine

- ABO = Approved by the Owner

PUMP DESCRIPTION (Cont'd)

All the following data shall be submitted by the Bidder for each metering pump but not limited to:

SPECIFICATION SCOPE		ITEM 3	ITEM 4
3.1	Equipment Name.....	Sodium hypochlorite metering pumps	Hydrazine metering pump
3.2	Quantity.....	2 x 100%	2 x 100%
3.3	Location.....	Water treatment area	Steam Turbine Building
3.4	Liquid to be handled.....	Sodium Hypochlorite	Hydrazine
3.5	Temp. of liquid.....(°C)	20 – 40	20 – 40
3.6	Pump suction from.....	Water supply sodium Hypochlorite solution tank	Hydrazine tank
3.7	Pump discharge to.....	To the discharge of supply water line	Condensate deaerator drum
3.8	Pump capacity.....(l/s)	By Bidder -ABO	By Bidder -ABO
3.9	Pump suction head.....(m)	By Bidder -ABO	By Bidder -ABO
3.10	Total head.....(m)	By Bidder -ABO	By Bidder -ABO
3.11	Material*		
3.11.1	Casing.....	-CD -ABO	-CD -ABO
3.11.2	Diaphragms.....	-CD -ABO	-CD -ABO
3.12	Flange		
3.12.1	Suction flange rating.....	By Bidder -ABO	By Bidder -ABO
3.12.2	Discharge flange rating (ANSI std)	By Bidder -ABO	By Bidder -ABO

PUMP DESCRIPTION (Cont'd)

All the following data shall be submitted by the Bidder for each metering pump but not limited to:

SPECIFICATION SCOPE		ITEM 5	ITEM 6
3.1	Equipment Name.....	Ammonium hydroxide metering pumps	Phosphate metering pumps
3.2	Quantity.....	2 x 100%	2 x 100%
3.3	Location.....	Steam Turbine Building	Steam Turbine Building
3.4	Liquid to be handled.....	Ammonium hydroxide	Phosphate
3.5	Temp. of liquid.....(°C)	20 – 40	20 – 40
3.6	Pump suction from.....	Ammonium hydroxide tank	Phosphate tank
3.7	Pump discharge to.....	To upstream of condensate pumps	Boiler steam drum
3.8	Pump capacity.....(l/s)	By Bidder -ABO	By Bidder -ABO
3.9	Pump suction head.....(m)	By Bidder -ABO	By Bidder -ABO
3.10	Total head.....(m)	By Bidder -ABO	By Bidder -ABO
3.11	Material*		
3.11.1	Casing.....	-CD -ABO	-CD -ABO
3.11.2	Diaphragms.....	By Bidder -ABO	By Bidder -ABO
3.12	Flange		
3.12.1	Suction flange rating.....	By Bidder -ABO	By Bidder -ABO
3.12.2	Discharge flange rating (ANSI std)	By Bidder -ABO	By Bidder -ABO

PUMP DESCRIPTION (Cont'd)

All the following data shall be submitted by the Bidder for each metering pump but not limited to:

SPECIFICATION SCOPE		ITEM 7	ITEM 8 (N/A)
3.1	Equipment Name..... :	Conc. Hydrazine metering pumps	Waste water acid pH control metering pumps
3.2	Quantity..... :	1 x100%	2 x100%
3.3	Location..... :	Steam Turbine area	Waste water, Treatment area
3.4	Liquid to be handled..... :	Conc. Hydrazine	Hydrochloric acid
3.5	Temp. of liquid.....(°C) :	20 – 40	20 – 40
3.6	Pump suction from..... :	Conc. Hydrazine Tank	Acid storage tank
3.7	Pump discharge to..... :	Condensate deaerator HRSG drum	pH control pit
3.8	Pump capacity.....(l/s) :	By Bidder -ABO	By Bidder -ABO
3.9	Pump suction head.....(m) :	By Bidder -ABO	By Bidder -ABO
3.10	Total head.....(m) :	By Bidder -ABO	By Bidder -ABO
3.11	Material*		
3.11.1	Casing..... :	-CD -ABO	-CD -ABO
3.11.2	Diaphragms..... :	By Bidder -ABO	By Bidder -ABO
3.12	Flange		
3.12.1	Suction flange rating..... :	By Bidder -ABO	By Bidder -ABO
3.12.2	Discharge flange rating (ANSI std) :	By Bidder -ABO	By Bidder -ABO

PUMP DESCRIPTION (Cont'd)

All the following data shall be submitted by the Bidder for each metering pump but not limited to:

SPECIFICATION SCOPE		ITEM 9 (N/A)	ITEM 10 (N/A)
3.1	Equipment Name..... :	Waste water caustic pH control metering pumps	Waste water coagulant metering pumps
3.2	Quantity..... :	2 x 100%	2 x 100%
3.3	Location..... :	Waste water, Treatment area	Waste water, Treatment area
3.4	Liquid to be handled..... :	Sodium Hydroxide	Coagulant
3.5	Temp. of liquid.....(°C) :	20 – 40	20 – 40
3.6	Pump suction from..... :	Caustic storage tank	Coagulant tank
3.7	Pump discharge to..... :	pH control pit	Mixing pit
3.8	Pump capacity.....(l/s) :	By Bidder -ABO	By Bidder -ABO
3.9	Pump suction head.....(m) :	By Bidder -ABO	By Bidder -ABO
3.10	Total head.....(m) :	By Bidder -ABO	By Bidder -ABO
3.11	Material*		
3.11.1	Casing..... :	-CD -ABO	-CD -ABO
3.11.2	Diaphragms..... :	By Bidder -ABO	By Bidder -ABO
3.12	Flange		
3.12.1	Suction flange rating..... :	By Bidder -ABO	By Bidder -ABO
3.12.2	Discharge flange rating (ANSI std) :	By Bidder -ABO	By Bidder -ABO

PUMP DESCRIPTION (Cont'd)

All the following data shall be submitted by the Bidder for each metering pump but not limited to:

SPECIFICATION SCOPE		ITEM 11(N/A)	ITEM 12 (
3.1	Equipment Name..... :	Waste water coagulant aid metering pumps	Desalination chemical injection pump
3.2	Quantity	-CD -ABO	-CD -ABO
3.3	Location..... :	Waste water, Treatment area	Desalination plant area
3.4	Liquid to be handled..... :	Coagulant aid	Chemical
3.5	Temp. of liquid.....(°C) :	20 – 40	20 – 40
3.6	Pump suction from..... :	Coagulant aid tank	Chemical tank
3.7	Pump discharge to..... :	Mixing pit	Inlet pipe
3.8	Pump capacity.....(l/s) :	By Bidder -ABO	By Bidder -ABO
3.9	Pump suction head.....(m) :	By Bidder -ABO	By Bidder -ABO
3.10	Total head.....(m) :	By Bidder -ABO	By Bidder -ABO
3.11	Material*		
3.11.1	Casing..... :	-CD -ABO	-CD -ABO
3.11.2	Diaphragms..... :	By Bidder -ABO	By Bidder -ABO
3.12	Flange		
3.12.1	Suction flange rating..... :	By Bidder -ABO	By Bidder -ABO
3.12.2	Discharge flange rating (ANSI std) :	By Bidder -ABO	By Bidder -ABO

4. DETAILED REQUIREMENTS

4.1 General

Each pump and its appurtenances and accessories shall conform to the requirements of this Specification.

The Total Head (H) specified by the Bidder to the pump manufacturer shall be calculated based on commercial pipes (not new and clean) as indicated in Williams & Hazen's formula.

4.2 Pump Type and Design

All metering pumps shall be of the positive displacement type.

All metering pumps shall be sized based on supplying 100% of the demand flow rate plus a 10% margin.

Each metering pump shall be furnished complete with a motor, isolation valves, suction and discharge check valves, a discharge pulsation damper, internal and external relief valves, a local pressure indicator and backpressure valve. Each concentrated hydrochloric acid metering pump shall also be furnished with a combination sludge trap and glass wool cartridge filter. All other metering pumps shall be furnished with a Y-strainer and a calibration reservoir.

Suction piping shall be sized and designed to ensure compatibility with the NPSH requirements of the pumps.

Each pump and motor drive shall be mounted on a common bedplate.

Each pump shall operate without undue strain or wear and without damage to any part of the pumping unit. It shall be Bidder's responsibility to determine the exact type of pump design which will provide such service without operating difficulty or undue replacement of parts.

Bidder shall be responsible for determining that each pump and driver operate in dynamic balance as a unit without undue vibration.

4.3 Pump Characteristics

Each pump shall have accurate proportion capacity using stroke adjustment, suitable for the service specified. Each pump and motor combination shall have non overloading characteristics.

Duplicate pumps designed for the same conditions shall operate satisfactorily in parallel with each other parallel or with other pumps of similar characteristics.

The pump characteristics shall include NPSH required (NPSHR) curves based on “as tested” results based on a 0% head deviation at the “break away” points. The recommended minimum NPSHR for normal continuous operation shall be indicated on the curves, for all flows including maximum run out.

A minimum margin of 25% shall be maintained between NPSHR and NPSHA.

Before manufacture of a pump is started, the Bidder shall submit to the Owner for review the exact pump capacity and head conditions.

4.4 Material

All pump components in contact with the pumped fluid shall be replaceable without removal of any piping.

4.5 Casing

Pumps shall constructed to permit ease of access for inspection and removal of internals and replacement of wearing parts.

Pump suction and discharge connections shall be flanged unless otherwise specified.

4.6 Diaphragms

Each diaphragm shall accurately fix its bed.

Diaphragms shall be designed to withstand maximum flexing with stresses below the material endurance limit regardless of stroke length setting. They shall also be of sufficient thickness and density to prevent permeation.

4.7 Sub Rings (if applicable)

The design of rings shall be such as they shall be easily renewed and maintained in service and shall reduce leakage to a minimum.

4.8 Shaft

Pump shaft shall be forged stainless steel and shall be accurately finish ground. Shaft surface shall be hard chrome plated or equivalent except otherwise specified.

4.9 Head Flange (if applicable)

Each head flange shall be bored and accurately fitted and rigidly keyed to shaft.

4.10 Gear Box

A suitable gear reduction shall be provided in conjunction with various mechanisms such as eccentric, crank or cam for translating rotary motion into reciprocating motion.

4.11 Bearing and Lubrication

General Requirements

Bearings shall be located and arranged to facilitate alignment, maintenance and to permit easy replacement.

Bearings may be either grease or oil lubricated as approved. Sizes and types to be interchangeable with standard makes.

4.12 Couplings

A flexible shaft coupling with spacer shall be provided on all pumps, except close coupled types.

4.13 Bedplates (if applicable)

Each pump and driver, with the exception of motor- driven close-coupled pumps, shall be bolted to a common bedplate that shall extend under and beyond all parts of the pump and driver.

Pumps shall be properly dowelled to their bedplates.

Bedplates shall be of manufacturer's standard type with a continuous raised drip lip around all four sides. All outside surfaces of bedplates shall have a smooth and finished appearance.

4.14 Cleaning and Painting

Before shipment, all parts of the equipment shall be thoroughly cleaned of all mill scale, rust, grease and other foreign matter.

All external unmachined ferrous metal surfaces shall be painted in accordance with the requirement as mentioned in Part 4, Specification S-420 and S-430.

5. PERFORMANCE TESTING AND GUARANTEES

The Bidder shall conduct performance testing in accordance with Part 4, Subsection 2.6 Testing and Guarantees and shall guarantee the following system guarantees :

SITE TEST

After equipment is put in service, Bidder shall run a check test on the equipment to ascertain that it is performing properly in every respect. Testing shall be in accordance with the guidelines and requirements contained in the "Hydraulic Institute Standards". Any deficiencies brought out by this test shall be promptly corrected by Bidder at no cost to Owner.

6. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality Assurance/Quality Control shall conform to the requirements of Part 4, Subsection 2.7.

7. INSTRUMENTATION AND CONTROLS

Feed rate shall be effected by varying the stroke length.

It shall be possible to vary the stroke length whether the pump is running or not.

Variation in stroke length shall be such as to provide feed rate adjustment over at least a 10 to 1 range with better than 1 % accuracy.

All pumps shall have provision for manual variation and local indication of stroke length.

Where required by the application, provision shall be made to vary the stroke length as a function of a 4-20 mA DC signal without the use of instrument air.

8. ELECTRICAL SPECIFICATION

The Bidder shall furnish and install all electrical equipment and accessories required for metering pumps.

For general electrical requirements, see Part 4. Technical Specification E-001.

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